

As the receiver is clocked by RXCLK, not SYSCLK, a noisy clock or data line can cause some internal design constraints to be violated, putting the receiver core logic into undefined states. Make sure that the clock and data lines satisfy the Electrical Characteristics and timing requirements of the FSI module found in the device data sheet. Failure to do so can cause the receiver state machine to go into an unrecoverable error state. The receiver can only be recovered by undergoing a soft reset. To determine the state of the receiver core after an unexpected frame error, the application must check the receiver core status bit.

In addition to the above errors, buffer overrun or underrun can warrant a soft reset to resynchronize with the local application software. Refer to [Section 31.3.3.8](#) for more information on the receive buffers. The requirement of resetting the receiver due to overrun or underrun is up to the application.

After the receiver has been placed into soft reset, the application must notify the other device's transmitter to begin a new synchronization phase. The simplest way to achieve this is through a ping or error frame sent with a designated tag. If the application is not using the FSITX on the device with the detected error, some other method must be established. The other device must stop transmitting and begin a new synchronization phase.

31.3.3.11 FSI_RX Reset

The receiver module and the registers are reset by SYSRSn. The receiver core is reset by SYSRSn or by writing a 1 to RX_MASTER_CTRL.CORE_RST.

A module reset causes the registers to be reset to the default state. After a module reset, the receiver module must be re-initialized and the data link re-established.

31.3.4 Frame Format

The FSI module transmits and receives information in frames. Each frame contains multiple phases where different information can be found. The number of phases as well as the total length of the frame varies depending on the frame type being transmitted. Frames can be as short as 16-bits long for a ping or error frame or 288-bits long for a 16-word data frame.

In normal transmission mode, there are four preamble clock edges before the start of the frame and four post-frame clock edges (postamble). Data is transmitted on both edges of the clock (double data rate). The basic frame structure is shown in [Table 31-4](#). Each phase of the frame (such as start-of-frame, frame type, and so on) is transmitted with the most-significant bit first. [Table 31-4](#) describes the basic frame structure used by the FSI and adapted according to which frame type is transmitted.

Table 31-4. Basic Frame Structure

Idle State	Preamble	Start of Frame	Frame Type	User Data	Data Words	CRC Byte	Frame Tag	End of Frame	Postamble	Idle State
	1111	1001	4 bits	8 bits	1-16 words	8 bits	4 bits	0110	1111	

The FSI also supports a FSI-SPI compatibility mode. The SPI compatible frame structure is similar to a standard FSI frame, but there are differences. Refer to [Section 31.3.13](#) for more information on how to configure and use the FSI-SPI compatibility mode.

Note

One word of the FSI refers to 16 bits.

The terms “frame” and “packet” can be used interchangeably to describe the signaling format of the FSI.

31.3.4.1 FSI Frame Phases

The different phases of the frame structure are described in detail.

- **Idle State:** During the idle state, the clock and data lines are driven high, the inactive state.
- **Preamble:** The preamble phase contains four clock edges (or two complete clock pulses) with the data signals held in the high state. These clock edges serve to flush the receiver logic and prepare the receiver logic for receiving a new frame. This phase is not present in SPI compatibility mode.
- **Start of Frame:** The start of frame phase contains two clock pulses with four bits, 1001, transmitted on the data lines.
- **Frame Type:** The frame type phase contains two clock pulses with the 4-bit frame type code being transmitted on the data lines. The different frame types are described in detail in [Section 31.3.4.2](#). The transmitter must set the TX_FRAME_CTRL.FRAME_TYPE field before transmitting a frame. The received frame type is stored in the RX_FRAME_INFO.FRAME_TYPE.
- **User Data:** The user data phase contains a fully user-configurable data field. There are no restrictions on how this field is used. This phase is only available in data frames. The user data to be transmitted is set by writing to TX_FRAME_TAG_UDATA.USER_DATA. The received user data is stored in RX_FRAME_TAG_UDATA.USER_DATA.
- **Data:** The data phase contains the data that is being transmitted. The data is pulled from the transmit buffer of the transmitter and is placed in the receive buffer of the receiver. Word 0 is transmitted first. This phase is only present in data frames. Depending on the type of frame transmitted, this can contain anywhere between 1 and 16 words depending on the frame type selected. More information on data frames is found in [Section 31.3.4.2.3](#).
- **CRC Byte:** The CRC byte contains the CRC of the transmitted data. The value present in this phase can be sourced from either hardware or software based on the TX_OPER_CTRL_LO.SW_CRC bit. Refer to the module-specific section of the CRC Submodule for more information on the CRC is generated or used, for the transmitter and receiver modules respectively. The CRC byte is only present in data frames.
- **Frame Tag:** The frame tag contains the 4-bit user-defined frame tag. There are no restrictions on how this field is used in an application. The transmitter supplies this tag into the TX_FRAME_TAG_UDATA.FRAME_TAG bits for data frames. Ping frames use the tag defined in TX_PING_TAG.TAG. The receiver can access the received frame tag in RX_FRAME_TAG_UDATA.FRAME_TAG.
- **End of Frame:** The end of frame contains four clock edges with four bits, 0110, transmitted on the data lines.
- **Postamble:** The postamble contains four additional clock edges with the data lines held in the high state. After the postamble, the clock and data lines are driven high (inactive state). This phase is not present in FSI-SPI compatibility mode.

31.3.4.2 Frame Types

The FSI hardware can generate and handle many predefined frame types. The different frame types can be used by the application to signal different types of events or convey different information to the receiver. The different frame types influence which phases and data fields to include in the transmitted frames.

[Table 31-5](#) provides a short overview of the different frame types used by the FSI. Each frame type is described in more detail in the following subsections.

Table 31-5. Frame Types and the 4-bit Codes

Frame Type	4-bit Frame Code	Description
PING	0000	Used typically for checking line integrity. A ping frame can be sent either by software or automatically by hardware.
ERROR	1111	Used typically during error conditions or any condition where one side wants to signal the other side for attention. However, the user software can use an error frame for any purpose.
DATA_1_WORD	0100	1 word data packet (16 bits of data)
DATA_2_WORD	0101	2 word data packet (32 bits of data)
DATA_4_WORD	0110	4 word data packet (64 bits of data)
DATA_6_WORD	0111	6 word data packet (96 bits of data)
DATA_N_WORD	0011	N(1-16) word data packet where software has programmed the number of the data words in a designated register. Both transmitter and receiver modules must have the same value programmed.
Reserved	0001, 0010, and 1000-1110	Reserved

31.3.4.2.1 Ping Frames

Ping frames are one of the most basic frames that can be generated by the FSI. [Table 31-6](#) shows the structure of the ping frames.

Table 31-6. Ping Frame

Idle State	Preamble	SOF	Frame Type	Frame Tag	EOF	Postamble	Idle State
	1111	1001	0000	xxxx	0110	1111	

The ping frame type is always 0000. The frame tag is defined by the application. Separate frame tags exist for timer and software initiated ping frames. No data or CRC is transmitted in a ping frame.

The main purpose of the ping frame is to periodically send a notification to the receiver to make sure an active connection between the transmitter and receiver. The transmitter and receiver cores implement different features to allow the ping frame to operate as a line break detect feature.

On the transmitter, the ping frame is the only frame that can be set up and transmitted without any further software or DMA intervention. Ping frames can be transmitted by any (or all) of the three sources: automatic ping timer, software, or external triggers. See [Section 31.3.2.3.3](#) for information on how the transmitter configures and sends the ping frames.

The receiver has a ping watchdog that can detect if a ping frame has not been received in a predetermined window. This allows the receiver to know if the connection between the receiver and the transmitter has been broken. See [Section 31.3.3.4](#) for information on how the receiver handles ping frames.

31.3.4.2.2 Error Frames

Error frames are similar to ping frames in that there are no data fields transmitted. Despite the naming of this frame as an “error frame,” the usage of it is up to the application, as no restrictions are placed on how and when this type of frame is transmitted. [Table 31-7](#) shows the structure of an error frame.

Table 31-7. Error Frame

Idle State	Preamble	SOF	Frame Type	Frame Tag	EOF	Postamble	Idle State
	1111	1001	1111	xxxx	0110	1111	

The structure of the error frame is the same as a ping frame. No data or CRC values are transmitted. The frame type is 1111 for all error frames, and the frame tag is defined by software in the TX_FRAME_TAG_UDATA register.

The receiver can detect if an error frame has been received based on the frame type field. Because of this, the receiver can read the incoming frame tag from the RX_FRAME_TAG_UDATA register and act on up to 16 different conditions.

31.3.4.2.3 Data Frames

Data frames are the most complex frames. As the name indicates, these frames are used to transfer data. [Table 31-8](#) shows the general structure of data frames.

Table 31-8. Data Frame

Idle State	Preamble	SOF	Frame Type	User Data	Data Words	CRC Byte	Frame Tag	EOF	Postamble	Idle State
	1111	1001	0xxx	xxxx xxxx	1-16 words	xxxx xxxx	xxxx	0110	1111	

The frame type field reflects the 4-bit code of the frame type. A list of frame types can be seen in [Table 31-5](#). The number of the data words transmitted is determined by the frame type chosen.

There are four fixed-length data frames supported by the frame type: 1 word, 2 words, 4 words, and 6 words.

Additionally, there is a user-defined data length frame type where the number of data words is fixed by software. Anywhere from 1 to 16 words can be transmitted in this frame type. This length must be configured in the N_WORDS field of the transmitter's TX_FRAME_CTRL register and receiver's RX_OPER_CTRL register.

31.3.4.3 Multi-Lane Transmission

The FSI is capable of transmitting and receiving data on two parallel data lines. When enabled, data bits are split between the data lines while the start of frame, frame type, frame tag, and end of frame fields are identical and complete on each line. The user data, data, and CRC fields are split between the data lines. Starting with the most-significant bit, the odd-numbered bits appear on D0 and even-numbered bits appear on D1.

In the following example, assume the following:

8-bit user data: u7u6u5u4u3u2u1u0

16-bit data: d15d14d13d12...d1d0

8-bit CRC: c7c6c5c4c3c2c1c0

Table 31-9. Multi-Lane Frame Format

Idle State	Preamble	SOF	Frame Type	User Data	Data Words	CRC Byte	Frame Tag	EOF	Postamble	Idle State
TXD0	1111	1001	0011	u7u5u3u1	d15d13...d1	c7c5c3c1	xxxx	0110	1111	
TXD1	1111	1001	0011	u6u4u2u0	d14d12...d0	c6c4c2c0	xxxx	0110	1111	

31.3.5 Flush Sequence

Every time there is a soft reset of the receiver, the receiver requires a flush sequence from the transmitter before the receiver can receive and decode frames. The receiver core has an asynchronous reset mechanism that allows the receive module to be reset even in the absence of the receive clocks. However, due to the design, this reset is released synchronous to the receive clock (RXCLK). Thus, the receiver requires five full clock pulses to be able to come out of reset. Sending the flush pattern makes sure that these clock edges are received and any subsequent frames sent to the receiver are correctly interpreted.

The flush sequence consists of a single toggle on both of the data lines as well as five consecutive pulses on the clock line.

If the FSI receiver is receiving data from a standard SPI, a data word of 0xFFFF from the SPI has the same effect as a flush sequence.

Figure 31-8 shows a sample plot of the flush sequence.

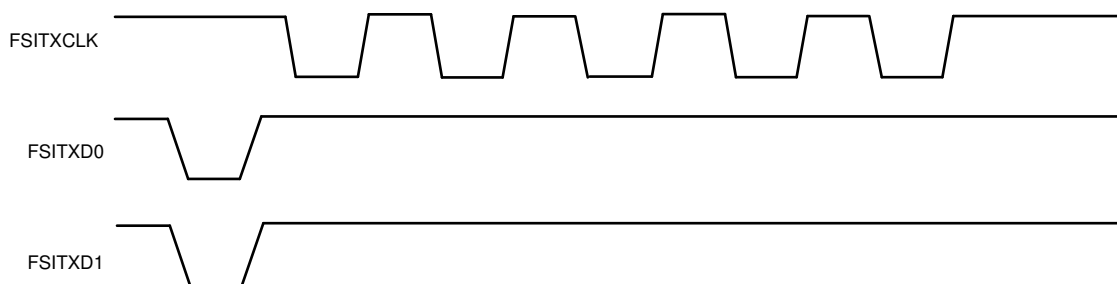


Figure 31-8. Flush Sequence Signals

31.3.6 Internal Loopback

The transmitter and receiver cores can be connected together internally to allow for development and debug. This is achieved by setting RX_MASTER_CTRL.INT_LOOPBACK to 1. Internal loopback routes the signals from the corresponding transmitter to the appropriate receiver pin. No configuration needs to be done in the transmitter.

Figure 31-9 shows the signal connections with internal loopback.

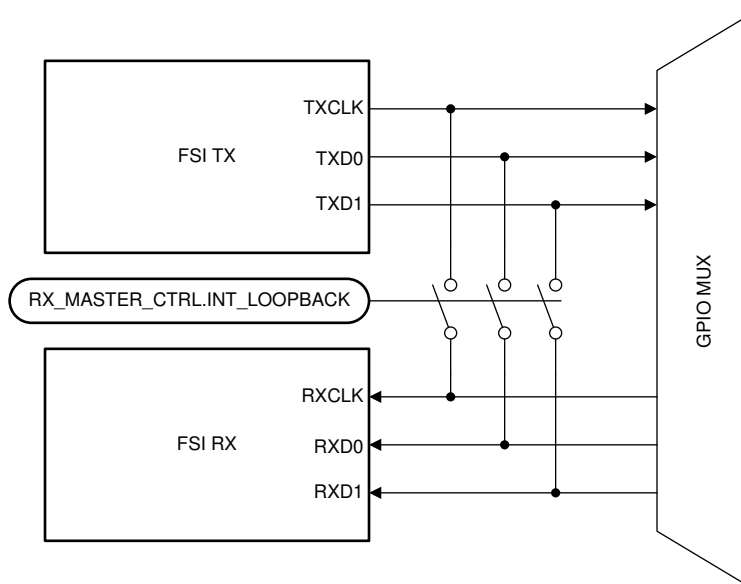


Figure 31-9. FSI with Internal Loopback